

Lecture 10

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E1 248
Sliding mode control and its application

1 Physical Interpretation of the Equivalent Control

2 Filippov Theory

Equivalent control method

Equivalent control method by the geometrical point of view

- Interpretation of the control signal in terms of its average or low-frequency behavior.
- The purpose of the control action is to ensure that the trajectories are driven towards and forced to remain on the sliding surface to guarantee a sliding motion.
- From a geometrical point of view, the equivalent control method does the same job. It replaces the discontinuous control on the intersection of the switching surface with a continuous one.

Mathematically, consider the system

$$\dot{x} = f(x) + B(x)u, \quad \text{where } x, f(x) \in \mathbb{R}^n, \quad B(x) \in \mathbb{R}, \quad u \in \mathbb{R}^n \quad (1)$$

Equivalent control method

Control input is defined as

$$u(x) = \begin{cases} u^+ & \text{if } s(x) > 0 \\ u^- & \text{if } s(x) < 0 \end{cases}$$

- Suppose at time t_s sliding surface is reached, and an ideal sliding motion takes place, i.e., $s(t) = 0 \forall t > t_s$. This implies $\dot{s}(t) = 0, \forall t \geq t_s$.
- Thus, the dynamics of the sliding variable is

$$\dot{s} = \frac{\partial s}{\partial x} \dot{x} = G(x)f(x) + G(x)B(x)u_{\text{equivalent}} = 0 \quad (2)$$

where $G(x) = \frac{\partial s}{\partial x}$.

- Assuming the matrix GB is non-singular for any x , find the equivalent control $u_{\text{equivalent}}$ as the solution of the Eqn.(2)

$$u_{\text{equivalent}} = -(G(x)B(x))^{-1}G(x)f(x) \text{ for } t \geq t_s.$$

- This is not the control signal which is actually applied to the plant but may be thought of as the control signal which is applied 'on average'.

Equivalent control method

Physical interpretation of the equivalent control

- For the occurrence of the ideal sliding mode, it was assumed that the control changes at a high (theoretically infinite) frequency such that the state vector is oriented precisely along the intersection of discontinuity surfaces.
- In reality, however, various imperfections make the state oscillate in some vicinity of the intersection, and control components are switched at finite frequency, alternatively taking the positive and negative values.
- These oscillations have high frequency as well as slow components.
- Almost all plants under control act as a low pass filter.
- Due to this low pass filter characteristic high frequency component is filtered out, and its motion in sliding mode is determined by the slow component.
- Practically, it is reasonable to assume that the equivalent control is close to the slow component of the real control, which can be derived by filtering out the high-frequency components using a low-pass filter.

Physical interpretation of the equivalent control

Mathematically, pass the discontinuous control signal through the low pass filter

$$\tau \dot{z} + z = u$$

to obtain the low frequency component $z(t)$.

The equivalent control is obtained as

$$\lim_{\tau \rightarrow 0} z = u_{\text{equivalent}}$$

where τ is the time constant of the low pass filter.

A brief review of Filippov theory

Convexification

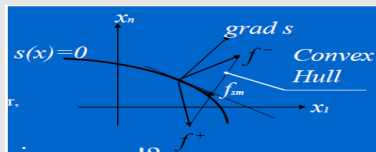


Figure: Filippov Solution

Physical interpretation of Filippov method using Regularization approach

Consider the system of the following form

$$\dot{x} = f(x, u), \quad x, f \in \mathbb{R}^n, \quad u(x) \in \mathbb{R}$$

$$u(x) = \begin{cases} u^+ & \text{if } s(x) > 0 \\ u^- & \text{if } s(x) < 0 \end{cases}$$

where the component of vector f , scalar functions u^+, u^- and $s(x)$ are continuous and smooth, and $u^+(x) \neq u^-(x)$.

A brief review of Filippov theory

Physical interpretation of Filippov method using Regularization approach

- Regularization theory studies methods for the solution to ill-posed problems (i.e., problems for which at least one of the conditions of uniqueness, the existence of the solution is not ensured)
- We assume that sliding mode occurs on the sliding surface $s(x) = 0$. Derive the motion equations using the regularization method.
- Let the discontinuous control be implemented. Control is assumed to take one of the two extreme values, u^+ or u^- , and the discontinuity points are isolated in time.
- Since discontinuity points are isolated in time, the solution exists in the conventional sense beyond the sliding surface.
- Further assume that the state velocity vector $f^+ = f_1 = f(x, u^+)$ and $f^- = f_2 = f(x, u^-)$ to be constant for some point x on the surface $s(x) = 0$ within a short interval $[t, t + \Delta t]$.

Physical interpretation of Filippov method using Regularization approach

Let the time interval Δt consists of two sets of intervals Δt_1 and Δt_2 such that $\Delta t = \Delta t_1 + \Delta t_2$, where Δt_1 and Δt_2 is the amount of that time when control of magnitude u^+ and u^- is active.

Mathematically increment of the state vector in this interval Δt is given by

$$\Delta x = f_1 \Delta t_1 + f_2 \Delta t_2$$

and the average state velocity is given as

$$\begin{aligned} \dot{x}_{\text{average}} &= \frac{\Delta x}{\Delta t} = \frac{f_1 \Delta t_1 + f_2 \Delta t_2}{\Delta t_1 + \Delta t_2} = f_1 \frac{\Delta t_1}{\Delta t_1 + \Delta t_2} + f_2 \frac{\Delta t_2}{\Delta t_1 + \Delta t_2} \\ &= \alpha f_1 + (1 - \alpha) f_2 \end{aligned}$$

where $\alpha = \frac{\Delta t_1}{\Delta t}$ is relative time for control to take value u^+ and $(1 - \alpha)$ to take value u^- and also $0 \leq \alpha < 1$.

Physical interpretation of Filippov method using Regularization approach

To get the velocity vector $\dot{x}$ along the sliding surface we have to take limit $\Delta t \rightarrow 0$. Hence sliding motion is represented as

$$\dot{x} = \alpha f_1 + (1 - \alpha) f_2 \quad (3)$$

Remark

One can also interpret Eqn. (3) as the velocity vector in the vicinity of a point on a discontinuous surface which is complemented by a minimal convex set, and the state velocity vector of the sliding motion belongs to this set.

- Because the state trajectories during sliding mode are in the sliding surface $s = 0$, the parameter α should be selected such that the state velocity vector of the system (3) is in the tangential plane to the sliding surface.

Physical interpretation of Filippov method using Regularization approach

Mathematically, one can write

$$\dot{s} = \nabla[s(x)].\dot{x} = \nabla[s(x)](\alpha f_1 + (1 - \alpha)f_2) = 0$$

where $\nabla[s(x)] = \left[\frac{\partial s}{\partial x_1} \cdots \frac{\partial s}{\partial x_n} \right]$. The solution of the above equation is given by

$$\alpha = \frac{\nabla(s).f_2}{\nabla(s).(f_2 - f_1)} \quad (4)$$

Substituting the α from Eqn.(4) to (3), one can get motion in sliding mode as

$$\dot{x} = f_{\text{sliding}} = \frac{\nabla(s).f_2}{\nabla(s).(f_2 - f_1)} f_1 - \frac{\nabla(s).f_1}{\nabla(s).(f_2 - f_1)} f_2$$

Remark

Sliding mode occurs in the surface $s(x) = 0$, therefore, the function s and $\dot{s}$ have different signs in the vicinity of the surface and

$$\dot{s}^+ = \nabla(s).f_1 < 0,$$

$$\dot{s}^- = \nabla(s).f_2 > 0$$

Check that $\dot{s} = \nabla(s)f_{\text{sliding}} = 0$ for the trajectories of the system (4) and show that they are confined to the switching surface $s(x) = 0$.

Thank You!